

The Calculus of Association: A Formula for Artificial General Intelligence

Meaning Compression, Frozen Embedders as Designable Measurement Instruments, Derived Data Abundance, and Teleological Constellation Training

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Abstract. This paper advances one claim: intelligence is the measurement and composition of associations. A dictionary cannot define a word with a word—each entry sends you to others, a merry-go-round of symbols that never bottoms out (Harnad’s symbol grounding problem)—yet it works, because meaning is carried by the *associations* between words, not the words: model a dictionary as a graph and $\approx 99\%$ is definable from a grounding kernel of $\approx 1\%$. So meaning is grounded association. Human intelligence runs this calculus on a biological substrate; *artificial general intelligence is the same calculus written down*, and building it is what this paper does. The instrument is a *frozen embedder*: trained on a corpus and frozen, it stops being a vectoriser and becomes a measurement device that reports where an input sits in that corpus’s web of associations. The lens is manufacturable—commission one for any axis a corpus exists for—so a panel of them is a designable basis for meaning, general because the instruments cover any domain and the cross-terms bridge across domains.

Three constructs follow, each formalised and running in production. *Derived Data Abundance*: n real inputs through N frozen, approximately-independent instruments yield up to $n(N + \binom{N}{2} + 1)$ structured signals, the $\binom{N}{2}$ cross-terms being associations *between* associations. *Meaning compression* is the per-input yield, complementary to the compression-as-intelligence tradition. *Teleological Constellation Training* reuses the same panel as a frozen alignment target with a per-output cosine guard G_{τ} . Together they give an operational account of definition: a term is the constellation the other instruments form at one instrument’s index—Gärdenfors’s object category, instantiated at scale with an information-theoretic differentiation contract (every instrument must add ≥ 0.05 bits about a real outcome; no pair may correlate above 0.6).

Three systems instantiate the panel at growing size: ClipCannon (video, $N = 7$), Context Graph (text and code, $N = 13$, up to 92 signals/input), and socialmedia2.com (civic matching, $N = 21$). Measured anchors: a voice reproduced at 0.961 mean WavLM speaker-similarity (encoder-matched; DNSMOS 3.93/3.93) and a published style model that holds character under prompt injection with an emergent zero-shot transfer to Golden-Age Spanish. The bound is stated plainly: the calculus computes only associations present in the chosen corpora, the $\binom{N}{2}$ abundance is an upper bound under approximate independence capped by the data-processing inequality, and the gain is reduced sample complexity realised up to the panel’s effective rank.

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1 Introduction

1.1 Intelligence is the calculus of association

Open a dictionary and try to find where meaning is stored. You cannot define a word with a word: every definition is built of other words, whose definitions are built of others, and the trip is a merry-go-round of symbols that never comes to rest on what anything means. Harnad [23] named this the *symbol grounding problem* and made it vivid with the Chinese/Chinese dictionary-go-round: with only a monolingual dictionary in a language you do not speak, you pass endlessly from one meaningless string to another. The words, in themselves, are empty. And yet a dictionary works. What carries the meaning is not the words but the *associations* between them—which word is invoked to define which. Vincent-Lamarre et al. [43] made this quantitative: model a dictionary as a directed graph of definitional links and roughly 99% of it can be defined, by association alone, from a grounding kernel of about 1% (a minimum feedback vertex set) that must be anchored outside language. The mental lexicon has the same shape: a small-world, scale-free association graph that grows by differentiation [41].

That is the thesis of this paper, stated at full strength. *Intelligence is the measurement and composition of associations*; everything else—search, planning, generation—is downstream of it. Human intelligence is this calculus run on a biological substrate; *artificial general intelligence is the same calculus written down*—a designable, grounded, mathematical form of it—and building that form is what this paper does. The word *general* is not a boast but a consequence: because the instruments are commissionable for any axis of association and the cross-terms bridge any pair of domains, the same machine spans video, code, and civic matching without retraining a new model per task. A single representation, like a single word, means nothing in isolation; meaning is the structured relation between representations, anchored at a small set of grounded points. It follows that a system which can systematically *measure* associations (project an object through many lenses), *count* them (the combinatorial cross-terms), *differentiate* them (keep only those that carry new grounded information), and *compose* them (name regions and gate against them) is running the engine of intelligence itself. The rest of this paper builds that engine, formalises it as a *calculus of association*, and measures it in three production systems. The one load-bearing caveat—association is circular unless it is grounded, and what grounds ours is contact with real outcomes—is stated in §2.8. §8 then shows that the field’s leading accounts of intelligence—compression, generalisation, world-modelling, and breadth-times-depth—are four

projections of this one mechanism, and Table 3 maps every capability the field asks of AGI to a construct within it.

1.2 The data wall is a symptom of counting the wrong thing

The same lens explains the crisis the field is most worried about. Frontier pre-training has nearly reached the ceiling of high-quality public text. Villalobos et al. [42] put the indexed human-text stock at roughly 300 trillion tokens and project exhaustion somewhere between 2026 and 2032, sooner under over-training regimes. Longpre et al. [30] watched a quarter of the high-quality C4 tokens become restricted in a single year. The most-discussed response is to train on generator-synthesised data, and that road is bounded: Shumailov et al. [39] show in *Nature* that recursive self-generation drives an irreversible distributional drift they call model collapse. The commercial consequence is a data-licensing race running to the hundreds of millions per partnership. So the standing responses divide cleanly. Path 1 buys more real data. Path 2 makes more synthetic data. Both count Shannon-compressed tokens, and both stay inside the collapse-prone recursion the moment they generate. The wall is what you hit when you count tokens instead of the associations among them; this paper counts the associations.

1.3 The reframe: a frozen embedder is a designable lens of association

This paper takes a third path, and it starts by asking what an embedder actually *is*. Training an embedder on a corpus and then freezing its weights does not store the corpus. It stores the geometry of how everything in that corpus related to everything else. The frozen weights are the associational structure of the training data. So when a new input x is projected through a frozen ϕ , the output $\phi(x)$ is not an encoding of x at all. It is a measurement of x against that whole web of associations, returning x ’s coordinates inside it.

Three moves stack on this observation, and they organise the paper.

1. *One frozen embedder is one designable lens of association.* The corpus you train on is the lens you get. So the instrument is manufacturable. A “legal-risk” lens is a model frozen on legal text; an “affective-resonance” lens is one frozen on annotated affect. You are not stuck with off-the-shelf semantics. You can commission an instrument for any axis of association you can assemble a corpus for. The lens is designed by choosing what to remember.

2. *N lenses are N independent association-spaces, and the $\binom{N}{2}$ cross-terms are associations between associations.* The pairwise term does not measure the object twice. It measures how the object’s position in lens A relates to its position in lens B. That is a structured-analogy operator, and it can relate two things that share no surface features at all, because the relation lives in latent, transitive association.
3. *A centroid plus a threshold is a label; a centroid set plus its cross-terms is a definition.* Name a region of an association-space (a constellation centroid), gate membership with G_τ , and continuous geometry becomes a discrete, countable, auditable label—meaning compression in one line. Add the cross-terms and the label hardens into a *definition*: the position in any one lens is only an index, and the meaning is the constellation of positions the other lenses report at that index (§2.5).

Section 2 develops the reframe and its consequences in full; the rest of the paper formalises and measures them.

1.4 The three constructs and the contribution

Derived Data Abundance (DDA) is the principle that a fixed corpus of n inputs, projected through a panel of N approximately-independent frozen instruments, yields $n \cdot (N + \binom{N}{2} + 1)$ structured supervisory signals at the dataset level for a code substrate, which works out to as many as 92 per input at $N = 13$ once a chunk-foundationality scalar is included. Meaning compression is the ratio view of that same fact—structured-signal-yield per input—which I situate as a complementary measure within the compression-as-intelligence tradition rather than a rival taxonomic category. Teleological Constellation Training (TCT) is the three-phase method that builds a frozen multi-modal centroid set C_P , trains a generator against it with a geometric-proximity loss, and enforces a per-output cosine guard G_τ at inference. The same frozen panel plays three roles in TCT (retriever, labeller, runtime guard) with no parameter updates.

So the contribution is one architectural move, three consequences, and one reframe. The move is to use a single frozen embedder panel as retriever, labeller, and runtime guard at once, over identical parameter instances. The consequences are a counting identity ($|D'| = n \cdot (N + \binom{N}{2} + 1)$, up to 92 at $N = 13$; the multi-view interaction counted explicitly, not a new quantity), a constructive sample-complexity bound ($\mathcal{T}_{\text{TCT}}(P) = O(M \cdot k)$), and a per-output verifiability property that scalar-reward alignment does not provide.

The reframe is that all three are best read as a calculus of association over manufacturable measurement instruments. That reading is what makes the method general across video, code, and civic matching, and what ties it to the information-physics tradition in which information is a physical, countable substrate. Pushed one step further it is a theory of *definition* (§2.5): a term is the constellation the other instruments form at a given lens’s index, taken over its reference set—an operational, relational realisation of Gärdenfors’s geometry of concepts (§3).

The claim is precise. A frozen embedder panel used as retriever, labeller, and runtime guard admits a counting identity with a constructive upper bound, a sample-complexity theorem with measured operating points, and a per-output verifiability property. All of it is bounded by the associations present in the chosen corpora, and all of it is auditable. The counting identity bounds extractable *structure*, not information: by the data-processing inequality (§4.2) the panel cannot exceed the corpus’s information about any real outcome, so DDA’s operative benefit is reduced sample complexity, and the realised yield tracks the panel’s effective rank rather than the nominal $\binom{N}{2}$.

2 The Reframe: Frozen Embedders as a Calculus of Association

This section is the conceptual heart of the paper. It states what the later sections formalise.

2.1 An embedder is a frozen crystallisation of its corpus’s associations

Let ϕ be trained on corpus D_ϕ and then frozen ($\partial\phi/\partial\theta = 0$). The learned geometry of ϕ encodes the relational statistics of D_ϕ : which things co-occur, entail, resemble, precede, or contrast with which, at whatever level of abstraction the training objective rewarded. Freezing converts a learner into an instrument, a fixed function whose output coordinates report a new input’s position relative to the associational structure ϕ memorised. Because the instrument is deterministic and frozen, every later measurement is reproducible and auditable. That is the precondition for treating $\phi(x)$ as a supervisory signal rather than a transient activation.

2.2 The lens is manufacturable

Because D_ϕ determines the lens, the lens is a design choice. This is the most under-exploited fact in representation learning. Practitioners treat embedders

as fixed commodities, but they are commissionable measuring devices. Want an instrument that measures “narrative shape”? Train a classifier-embedding on story-structure annotations and freeze it (socialmedia2.com slot S5). Want to separate “causal want” from “causal give”? Train an asymmetric causal embedder on contribution statements and freeze the cause and effect halves (socialmedia2.com S3/S4). Want “code foundationality”? Compute a PageRank over a frozen dependency graph (Context Graph ψ_{found}). Assembled this way, the panel is not a model. It is a library of instruments, built per problem.

2.3 N lenses and the $\binom{N}{2}$ cross-terms: associations between associations

Project an object through N instruments and you get N coordinate vectors, one per association-space. The pairwise interaction features $\rho_{jk}(\phi_j(x), \phi_k(x))$ (a normalised cosine, a pairwise mutual-information estimate, or a concatenation through a frozen adapter) measure how an object’s position in space j relates to its position in space k . That is the structured-analogy operator, and it surfaces relations no single lens can see. The same operator underlies code-chunk and failure-mode, patch and nearest-precedent, person and gathering-format, voice-sample and identity: one mechanism across every domain. Its cardinality is $\binom{N}{2}$, which is why adding one instrument multiplies value against every existing instrument, and also why redundancy has to be policed (§2.8, §5.3).

2.4 Centroid plus threshold is a label (meaning compression)

A frozen centroid $c_m = \overline{\phi_m(R_m)} / \|\cdot\|$ names a region of association-space m , the region occupied by a reference set R_m (a subject, a style, an outcome class). The predicate $\mathbb{1}[\cos(\phi_m(x), c_m) \geq \tau_m]$ turns continuous geometry into a discrete, countable, auditable label: does x belong to the association-region of R_m ? Stack such labels across the panel and a fixed, unlabelled corpus becomes a dense, structured supervisory dataset; that is the static side of meaning compression. Gate generation against the same labels and alignment becomes checkable geometry; that is the dynamic side.

2.5 From label to definition: meaning is the cross-space binding

A single instrument, on its own, does not return a meaning. It returns an *index*: a coordinate $\phi_m(x)$ that says *where* x sits in lens m , not *what* sitting there amounts to. The meaning of that coordinate is supplied by the rest of the panel—by the constellation of positions

$\{\phi_k(x)\}_{k \neq m}$ the other instruments report when this one reports $\phi_m(x)$. This holds whichever instrument is taken as the anchor, so meaning is fully relational: no lens carries it alone, and the panel carries it in the bindings between lenses. The marginal positions are the signs; the cross-terms are the signified.

Make this precise at two levels. For a single input (the token level), its meaning under the panel is the relational object

$$\mathcal{M}(x) = \{ \phi_m(x) \mapsto \{ \phi_k(x) \}_{k \neq m} : m = 1..N \}, \quad (3)$$

equivalently fixed by the joint coordinate together with all $\binom{N}{2}$ pairwise bindings $\rho_{jk}(\phi_j(x), \phi_k(x))$. For a term (the type level), its *definition* is the characteristic such object over a reference set R_t ,

$$\text{Def}(t) = \left(\{ (c_m, \tau_m) \}_{m=1}^N, \{ \rho_{jk} \}_{j < k} \right), \quad (4)$$

the per-lens centroid-threshold regions (the marginals, indexical) *and* the characteristic cross-lens correlations (the bindings, constitutive). An input instantiates t when its constellation falls inside the marginal regions *and* reproduces the characteristic bindings. The marginal regions alone are a list of coordinates; the bindings are what make the list a concept. This is why the $\binom{N}{2}$ cross-terms are not a bookkeeping convenience: they are the part of the panel that carries meaning, and the same quantity that drives DDA’s signal yield (§4.2) is what turns a constellation into a definition.

Worked example (“happy”). Take Clip-Cannon’s panel and anchor on the affect instrument (`wav2vec2-large`, a dimensional arousal/valence/energy space). A high-valence coordinate is only an index. What *defines* the input as happy is the constellation at that index: the prosody register, the visual expression, the semantic content, *and*, decisively, the bindings among them. Genuine happiness, sarcasm, and nervous laughter can share the affect marginal yet differ in the affect $\leftrightarrow$ prosody and affect $\leftrightarrow$ visual cross-terms; the marginal cannot separate them and the bindings can. “Happy” is therefore not a point in affect-space but a characteristic cross-space signature, and the same construction names a term by anchoring on *any* instrument and reading the constellation the others form. This is, formally, what Gärdenfors [20] calls an object category and what operationalism [6] calls a concept fixed by its measurement operations (§3); its boundary is stated in §2.8.

This is also where the thesis of §1.1 becomes mechanical. A dictionary defines a word by the constellation of other words its definition invokes; $\text{Def}(t)$ defines a term by the constellation of other instruments’ coordinates at its index. They are the same object—meaning as structured association, not as any single symbol—and

Def(t) is its machine-computable form. The operations the thesis names are now explicit: *measure* is the projection $\{\phi_m(x)\}$, *count* is the $\binom{N}{2}$ cross-terms, *differentiate* is the bits-against-a-real-outcome contract (§5.3), and *compose* is the centroid-and-threshold predicate. Run all four over a panel anchored to real outcomes and you are computing meaning rather than retrieving it; that is the sense in which the calculus of association is a formula for artificial general intelligence—the mathematical form of the same association-engine that human intelligence runs biologically, made general by designable instruments and domain-bridging cross-terms. What keeps it from being the empty dictionary-go-round is grounding, treated next.

2.6 What this makes possible

- A designable measurement basis for meaning. You build the exact association-axes a problem needs by choosing corpora and freezing, and each new axis earns its place only by adding differentiated information (§5.3).
- Structured supervision from fixed, unlabelled data at $N(N-1)/2$ density, without generating anything, which places it outside the model-collapse recursion (§4.5).
- Cross-domain bridging and analogy at scale, through the transitive cross-terms (§2).
- Per-output, model-free verification. Frozen and deterministic means membership is reproducible and auditable, so alignment becomes geometry you can check rather than a scalar you trust (§6).
- Identity-locked, controllable generation. Pin a generator to a named region and it cannot drift out without failing the gate, whether the target is a voice, a style, a face, or a persona.
- Field mapping. Project a domain’s corpus through the panel and you get a labelled association-graph of that field (Context Graph’s chunk–skill–ability graph; *socialmedia2.com*’s 21-axis citizen manifold).
- Composability and transfer. Train once, reuse forever. The marginal cost of a new axis is one frozen forward pass and its marginal value is its new bits, which is a different economics from per-task retraining.

2.7 Query-adaptive weighting: the agent steers the panel

The panel is not searched with a fixed metric. Because every instrument’s axis of association is named and meaningful, an agent holding a query can assign

each instrument a *query-specific* weight and search the weighted combination,

$$\text{score}(x | q) = \sum_{m=1}^N w_m(q) \text{sim}_m(\phi_m(x), \phi_m(q)), \quad (6)$$

fused over the per-instrument rankings (weighted reciprocal-rank fusion in the running systems). One query might weight the lexical and code instruments heavily and the temporal ones at zero; the next might invert that—the same panel, tilted differently. The weight vector lives on the $(N-1)$ -simplex (and, once cross-term weights are admitted, a larger space), so each query selects a point in a continuous control space whose dimension *grows with the number of instruments*. More lenses therefore mean more degrees of freedom to express what to look for, and retrieval sharpens accordingly: the agent does not merely read the constellation, it re-weights it toward the query. This is Gärdenfors’s context-dependent salience—categorisation under weighted quality dimensions—made an explicit, per-query operation. The gain is real only up to the panel’s differentiated rank: reweighting redundant axes adds no information (§4.2), so the differentiation contract (§5.3) and query-adaptive weighting are two sides of one design—every instrument that earns its place by adding bits also adds a usable axis of control.

2.8 The boundary that keeps the claim precise

The claim is bounded, and the bound is what makes it strong. Several guardrails keep it precise, and the first is the one the intelligence claim of §1.1 stands or falls on.

- **Grounding, or the dictionary-go-round returns.** Association alone is circular: a panel of lenses defining objects only by reference to one another is exactly Harnad’s monolingual dictionary, an endless pass from symbol to symbol that never reaches meaning [23]. The calculus escapes the circle only because it is *anchored*: the associations are measured against, and admitted by their bits about, a real non-linguistic outcome—the Docker oracle’s pass/fail in Context Graph, the recorded speaker in voice cloning, the 30-day in-person tie-firmation outcome at *socialmedia2.com*. That anchor set is the analogue of the dictionary’s $\approx 1\%$ grounding kernel [43]: small, but the thing that makes the other 99% mean anything. “Intelligence is association” is therefore precise only as “intelligence is *grounded* association”; strip the anchor and the formula describes a very efficient way of computing nothing. The data-processing bound (§4.2) is the formal statement of this ceiling.

- It computes associations that are present in the training corpora, and those are measurable, bounded, and auditable. The choice of corpus is the choice of instrument, which is why social-media2.com runs a quarterly redundancy and retirement audit (§5.3) and keeps a confounder and proxy-risk register.
- The $\binom{N}{2}$ abundance is a constructive upper bound under approximate independence. Two lenses that encode the same associations add few bits, so the differentiation contract reads “associations between anything, minus the redundant ones, measured in bits against a real outcome.”
- Cross-lens comparison stays structured. The relationships are scalar: mutual information and agreement, never dim-wise flattening. The associations live between the spaces, and the panel is never collapsed into a single vector.
- **Representation, not extrapolation.** To measure anything true about a target Y , the corpus must contain observations of Y and the labels must come from Y . The instrument interpolates *within* the distribution its corpus represents; it cannot extrapolate to a phenomenon with zero representation in the training data. An axis the corpus never saw is an axis the panel cannot report, however many cross-terms are formed. This is the boundary that keeps the calculus empirical: it describes the associational structure of assembled corpora, and claims nothing about phenomena outside them.
- **Definition is operational and relational, not essential.** A term is fixed by where its reference set measures and how those measurements co-vary, relative to the panel and corpus—an extensional, operational definition in Bridgman’s sense [6], not a statement of necessary-and-sufficient essence. Change the panel and the definition changes, and the data-processing bound (§4.2) caps it at the corpus’s information about any real outcome. The cross-terms are well-posed as meaning only where each instrument’s training objective made the target property a coherent direction on its sphere (§6.5); where it did not, they degrade to noise and the differentiation contract (§5.3) is what catches it.

3 Related Work (condensed)

Data wall and collapse. Villalobos et al. [42] and Longpre et al. [30] frame the stock-and-accessibility problem. Shumailov et al. [39] (*Nature*), Dohmatob et al. [15], and Alemohammad et al. [1] establish irreversible drift under the recursion $D_{t+1} \ni \tilde{x} \sim p_{\theta_t}$. The

two cited mitigations, Gerstgrasser et al. [22] (accumulation) and Feng et al. [17] (verification), both stay inside that recursion. DDA’s signals are frozen measurements of real inputs rather than generator samples, so the recursion precondition does not hold (§4.5).

Compression as intelligence. Delétang et al. [13] run LLMs as general compressors, and Huang et al. [24] report a correlation of about 0.95 between compression ratio and capability across more than thirty LLMs. The standard taxonomy of bit-, weight-, and activation-compression measures economy of output relative to a same-modality input. None of it measures labelled-signal yield per input. I propose meaning compression as a fourth axis (§4.3), an extension of this line rather than a contradiction of it.

Multi-embedder retrieval. SigLIP [46], Nomic Embed [33], WavLM [7], ColBERTv2 [38], SPLADE [19], and KEPLER [44], with HNSW [31] and RRF [10], are the standard parts. That literature uses N embedders as retrieval instruments, scored by NDCG, Recall@k, or EER. None uses the same N as a labeller over a fixed corpus and a per-output runtime guard at the same time. The role-flip over identical frozen instances is the novelty relative to this pool.

Multi-view learning and co-training. The closest theoretical ancestor of DDA is multi-view learning. Blum and Mitchell [4] prove that when two views are conditionally independent given the label—“ x_2 is conditionally independent of x_1 given the classification”—“any weak hypothesis can be boosted from unlabeled data,” and Dasgupta et al. [11] give the corresponding PAC generalization bounds in terms of inter-view agreement. Sridharan and Kakade [40] recast the assumption information-theoretically and derive a “normative justification for canonical correlation analysis (CCA)” under it. DDA’s “approximately-independent instruments” assumption is exactly this conditional-independence condition, and the $\binom{N}{2}$ cross-terms are the multi-view interaction the same tradition studies; the counting identity (1) is a bookkeeping of that interaction, not a new quantity. What is new here is not the combinatorics but two engineering moves: (i) the views are *commissioned and frozen foundation-model panels*—designed by choosing a training corpus—rather than a fixed feature split given by the data, and (ii) the same frozen panel is reused as retriever, labeller, and runtime guard over identical parameter instances. DDA is best read as multi-view learning carried onto designable, frozen embedder panels.

Representation convergence. The reframe also rests on an empirical regularity: Huh et al. [25] argue that “neural networks, trained with different objectives on different data and modalities, are converging to a shared statistical model of reality in their representation spaces,” with the converged kernel approaching

the pointwise mutual information over the latent events that generate observations. Two consequences matter here. First, it is what licenses a multi-instrument panel to be read as coordinates against one underlying association structure rather than N incommensurable spaces. Second, the cross-panel triangulation we use elsewhere (a disjoint Panel B re-encoding the same artifacts and reaching the same verdict) is, in effect, a local test of this convergence; where convergence fails, the cross-terms degrade to noise, which is precisely what the differentiation contract (§5.3) is built to catch.

Symbol grounding and the dictionary. The thesis (§1.1) rests on a named result. Harnad [23] posed the symbol grounding problem: a purely symbolic system—the monolingual dictionary-go-round—never reaches meaning, because symbols are defined only by other symbols. The escape is to ground a subset outside the symbol system. Vincent-Lamarre et al. [43] quantified the subset: as a directed graph, a dictionary reduces to a grounding kernel of $\approx 10\%$ and a minimum grounding set (feedback vertex set) of $\approx 1\%$ from which the rest is definable by association alone, and Steyvers and Tenenbaum [41] show the mental lexicon is itself a small-world, scale-free association graph grown by differentiation. We read these as the empirical backbone of “intelligence is grounded association”: association carries almost all of meaning, a small grounded anchor carries the rest, and our contribution is a system that makes both halves measurable—the associations as a frozen-instrument calculus, the anchor as contact with a real oracle outcome.

A geometric, operational, relational theory of meaning. The constellation-as-definition reading (§2.5) has deep roots outside machine learning; the contribution here is to instantiate them on frozen foundation-model panels at scale. Gärdenfors [20, 21] model concepts as regions in a space of quality dimensions and define an *object category* as a number of convex regions in different domains, “together with prominence values of the domains and information about how the regions in different domains are correlated”—domains, salience weights, and cross-domain correlations that map onto our frozen instruments, RRF slot weights, and $\binom{N}{2}$ cross-terms, and onto prototype theory’s centroids [37]. The relational half of the claim—that a single coordinate is only an index and meaning lives in the bindings—is structuralist [12] (“in language there are only differences without positive terms”), distributional [18] (“you shall know a word by the company it keeps,” here the company of other-instrument coordinates), and inferential-role [3] in spirit. Osgood et al. [34] already measured connotative meaning as a profile across many bipolar scales reduced by factor analysis; our panel is a learned, frozen, high-dimensional semantic differential. Defining a concept by the operations

that measure it is operationalism [6], whose known cost—instrument- and corpus-relativity—is exactly the boundary we state in §2.8 rather than evade. The discrete-feature limit is componential and compositional semantics [27, 45], and Du et al. [16] report that frozen multimodal models recover stable, interpretable concept dimensions aligned with human judgements— independent evidence that such panels carve meaning into nameable axes. None of this prior work composes a single frozen panel as retriever, labeller, and runtime guard over identical instances, nor enforces an information-theoretic differentiation contract (§5.3) over the cross-terms.

Identity-locked generation and alignment. Zero-shot voice cloning (VALL-E 2 at about 0.881 WavLM-SV [8]; NaturalSpeech 3 at about 0.891 [26]; the field clusters around 0.86 to 0.89) uses single-prompt conditioning, not a frozen multi-clip centroid with a per-output cosine gate. Scalar-reward alignment (InstructGPT [35], DPO [36], Constitutional AI [2]) optimises an expectation under a learned scalar and ships no per-output frozen-threshold predicate. The contrast with TCT is structural rather than a benchmark contest (§9.3). The closest geometric cousins are representation engineering [47], which places named directions in activation space at the centre of control, and activation steering—whose reliability Braun et al. [5] show is conditional: “vector steering is unreliable when the target behaviour is not represented by a coherent direction.” TCT inherits exactly this caveat, and §6.5 states when the cosine guard is and is not well-posed.

4 DDA and Meaning Compression

4.1 The DDA counting identity

Let $D = \{x_i\}_{i=1}^n$ be a fixed corpus and $\Phi = \{\phi_m\}_{m=1}^N$ a panel of frozen, approximately-independent instruments. For pairs $j < k$ let ρ_{jk} be a cross-embedder interaction feature, and for code chunks let $\psi_{\text{found}}(x) \in [0, 1]$ be the PageRank-derived foundationality scalar from the frozen dependency graph. The derived dataset materialises every per-embedder projection, every pairwise interaction, and the foundationality scalar:

$$|D'| = n \cdot \left(N + \binom{N}{2} + 1 \right), \quad (1)$$

which decomposes into nN per-embedder vectors, $n\binom{N}{2}$ pairwise interaction features, and n graph-foundationality scalars. At $N = 13$ the per-input yield is $13 + 78 + 1 = \mathbf{92}$; at $N = 7$ (no code graph) it is $7 + 21 = \mathbf{28}$; at $N = 21$ (no code graph) it is $21 + 210 = \mathbf{231}$.

Equation (1) is a structural upper bound under approximate independence, and the word *upper* is load-bearing. The nN term is tight as long as the instruments produce non-constant outputs, but the $\binom{N}{2}$ term is realised only under approximate orthogonality. In practice embedders trained on overlapping corpora are correlated, so the *effective* panel rank—the participation ratio $N_{\text{eff}} = (\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$ over the cross-embedder covariance spectrum $\{\lambda_i\}$ —is smaller than the nominal N , and the realised pairwise yield falls correspondingly below $\binom{N}{2}$. Pairwise decorrelation does not even certify the bound: three instruments can be pairwise uncorrelated yet jointly dependent, so the D1 contract (§5.3) bounds redundancy without certifying joint independence. The honest figures to report alongside the nominal count are therefore N_{eff} and the measured cumulative information gain $\sum_i \Delta H_i$, not $\binom{N}{2}$ taken as achieved. A signal here is a scalar or low-dimensional feature: a deterministic function of a real input or fixed-corpus graph under frozen instruments, attached to a specific index, usable as input or target without collecting new data. DDA is therefore not synthetic generation, not augmentation, and not distillation. The “approximate independence” invoked here is the conditional-independence condition of multi-view learning [4]: where it holds the cross-terms carry signal that no single lens sees, and where it fails the redundant pairs are bounded out by the differentiation contract (§5.3).

4.2 What the count is, and is not: the data-processing bound

Equation (1) counts *signals*, not *information*, and the distinction is the data-processing inequality. For any external target Y and any deterministic feature f of the frozen panel applied to the corpus,

$$I(D; Y) \geq I(f(\Phi(D)); Y), \quad (2)$$

with equality only when f is a sufficient statistic for Y . No marginal projection, no $\binom{N}{2}$ cross-term, and no foundationality scalar can carry more information about a real outcome than the raw corpus already held. DDA does not create information; it *re-exposes* latent information in a form a downstream learner can use. Its operative value is therefore *sample-complexity reduction*: by pre-structuring the manifold into up to $N + \binom{N}{2} + 1$ differentiated coordinates, a learner reaches a target outcome from far fewer labels than the raw input would demand, because structure that was implicit is now explicit and low-dimensional. “Abundance” in DDA names an abundance of *usable, pre-structured coordinates*, never an abundance of information beyond the $I(D; Y)$ that (2) caps. The counting identity is thus an upper bound on extractable structure, and the

information content of that structure is bounded by the corpus—the same boundary stated qualitatively in §2.8: a panel reports only associations its corpora contain.

4.3 Meaning compression: structured-signal-yield per input

The meaning-compression ratio is $\text{MC}(\Phi, D) = |D'|/\text{raw}(D) = N + \binom{N}{2} + 1$ per input. Conventional compression asks how few bits, parameters, or megabytes a fixed content can be squeezed into. Meaning compression flips the question: for a fixed amount of raw data, how many structured supervisory signals can you extract? The name is by analogy and is, strictly, a *yield* rather than a reduction—it multiplies usable coordinates rather than shrinking bits—so it is dual to, not a competitor of, the Kolmogorov/Shannon sense of compression that Delétang et al. [13] and Huang et al. [24] tie to capability. By the data-processing bound of §4.2 the underlying information is unchanged; what the yield measures is how much of it has been made explicit and low-dimensional. The measures work on genuinely different unit pairs, as Table 1 sets out.

The four are complementary axes, and a single pipeline can realise all of them. Meaning compression extends the conventional three rather than replacing any of them.

4.4 Foundationality-graded compression

One refinement carries over from the production code substrate. Foundationality-graded compression makes aggressiveness scale inversely with a chunk’s PageRank foundationality s_i , with $\alpha_{\text{compress}}(x_i) = \text{clip}(1 - 0.75 s_i, 0.25, 1.0)$. Foundational chunks are sampled $(1 + \lambda_{\text{found}} s_i)$ and Fisher-protected $(1 + \lambda_{\text{fisher}} s_i)$ more aggressively, yet compressed less, so the load-bearing structure of a codebase is preserved at higher fidelity than its periphery.

4.5 Scope of the collapse-silent claim

Collapse needs a generator-in-loop recursion $D_{t+1} \ni \tilde{x} \sim p_{\theta_t}$. DDA never instantiates one. Each $\phi_m(x) \in D'$ depends only on a real $x \in D$ and on frozen parameters, so D' is a decomposition of D rather than a sample from a generator. The precondition fails, and the cited theorems say nothing about this configuration. This is a scope argument, not an immunity theorem. Two boundary cases stay out of scope: inference-time best-of- K selection, which updates no parameter, and embedder-mediated data curation, which has a different feedback structure.

Table 1: Meaning compression as a complementary measure alongside the conventional three compression axes. The measures operate on different input/output unit pairs and are complementary rather than competing; meaning compression is a signal *yield*, dual to the bit-reduction sense.

Category	Input unit	Output unit	Canonical techniques
Bit	bits of raw data	bits of compressed data	gzip, FLAC, PNG, arithmetic coding
Weight	trainable parameters	accuracy at those parameters	GPTQ, SparseGPT, distillation
Activation	peak runtime memory	quality at that memory	TurboQuant, KV-cache quantisation
Meaning	one raw input	structured labelled signals	DDA (this paper)

4.6 The three-role composition

The same frozen Φ plays three roles at once over identical parameter instances, with no parameter update between or during them. As retriever it ranks stored content by cosine. As labeller it materialises the signals of (1). As runtime guard it evaluates a candidate against a frozen centroid (§6). Relative to the locked reference set, no cited paper composes all three roles together over identical frozen instances.

5 Three Production Systems

Three antecedent systems instantiate the labeller role at increasing panel size. All counts are system-at-date figures from the working tree.

5.1 ClipCannon: video, $N = 7$, up to 28 signals/input

ClipCannon is a $\approx 67,585$ -line Python package that runs a 23-stage analysis DAG over source video. It extracts seven modalities: visual (SigLIP-SO400M, 1,152-D), semantic (nomic-embed-text-v1.5, 768-D), emotion (wav2vec2-large, 1,024-D), speaker (WavLM-large, 512-D), prosody (custom 12-D), sentiment (MiniLM-L6-v2, 384-D), and voice identity (ECAPA-TDNN, 192-D). That is 4,044 dimensions across seven modalities, exposed as 58 MCP tools, with a SHA-256 tamper-evident provenance chain. At $N = 7$ it yields up to 28 signals/input. It is the labeller for Cases 2 and 3, and the curation substrate that turns one 16-minute interview into 2,362 training clips.

5.2 Context Graph: text and code, $N = 13$, up to 92 signals/input

Context Graph is a $\approx 370,000$ -line Rust workspace that embeds every stored memory through 13 frozen embedders (E1 to E13). Together they span 11,008 dense dimensions, two 30,522-term sparse vocabularies (E6 lexical, E13 SPLADE), and a 128-per-token late-interaction space (E12 ColBERTv2). The remaining slots are e5-

large-v2 semantic (E1), sinusoidal and Fourier temporal (E2 to E4), asymmetric Nomic causal (E5), Qodo 1,536-D code (E7), structural e5 for graph edges (E8), 10,000-bit HDC typo-robust (E9), paraphrase-asymmetric e5-base (E10), and KEPLER entity geometry (E11). E5, E8, and E10 store dual vectors (cause/effect; source/target; doc/query). Persistence is 59 RocksDB column families, and retrieval is weighted RRF over the per-embedder rankings, exposed as 75 MCP tools. At $N = 13$ it yields up to 92 signals/input ($13 + 78 + 1$). Context Graph is the backbone of Leapable.ai’s knowledge-vault product and the labeller for Case 1.

5.3 socialmedia2.com: civic matching, $N = 21$, and the differentiation contract

The most information-theoretically explicit instantiation is the civic-matching system behind socialmedia2.com, which matches citizens into durable real-world gatherings instead of optimising engagement. Its constellation is a 21-slot slate across 11 orthogonal axes plus a hard-constraint vector, roughly 35,800 dimensions per citizen, and it is never flattened. The axes and a few representative slots: semantic content (S1 **sem-self**, S2 **lex-vocab** SPLADE), causal want and give (S3/S4 asymmetric Nomic-causal), narrative form (S5 **story-shape**, S6 **hero-arc**), cognitive mode (S7 **reason-mode**, S8 **abstract-concrete**), moral foundations (S9 **mfq-vec**), ideological bridging (S10 **bridge-vec**), somatic and prosodic (S11 to S13 voice tempo, affect, energy), temporal rhythm (S14/S15 Fourier day and week), aesthetic field (S16 SigLIP-2 on curated images, S17 stylometry), network position (S18 TransE-class, S19 Burt structural-hole), and relational dynamics (S20 conflict-style, S21 care-mode). At $N = 21$ the DDA identity gives up to 231 signals/input ($21 + 210$).

socialmedia2.com turns the differentiation principle into an enforced engineering contract. A slate S is differentiated when:

- **(D1) Distinctness.** $\forall i \neq j : \text{corr}_+(E_i(X), E_j(X)) \leq 0.6$ on a calibration set $|X| \geq 500$.
- **(D2) Marginal information.** $\forall i : H(y | E_1..E_{i-1}) - H(y | E_1..E_i) \geq 0.05$ bits, where y is the binary 30-day in-person tie-formation outcome.
- **(D3) Frozen.** $\forall i, \forall \text{phase} \in \{\text{match, train, eval, serve}\} : \partial E_i / \partial \theta_i = 0$.

The principle behind the three conditions is that a panel’s value is $\sum_i \Delta H_i = \sum_i I(y; E_i | E_1..E_{i-1})$, not $\sum_i H(E_i)$. A redundant slot, with $\text{corr} \approx 1$, contributes $\Delta H_i \approx 0$ and is retired at the quarterly audit that runs over all $\binom{21}{2} = 210$ pairs. Instruments are admitted in bits against a real outcome, not by raw dimensionality. The slate registry, the 21-slot constellation assembly, slot-shape validation, the semantic probe suite, and the pairwise-redundancy report are implemented and operational; a five-answer intake produces a complete 21-slot constellation, and `assemble_constellation` succeeds only when all 21 slots are present. The ≥ 500 -citizen calibration corpus and the 30-day outcome validation that turn D2 into a measured pass are the launch milestone.

5.4 The three systems on one curve

Plot signals-per-input, $N + \binom{N}{2} (+1)$, against N and the three systems land on a single quadratic: ClipCannon at $N = 7 \rightarrow 28$, Context Graph at $N = 13 \rightarrow 92$, socialmedia2.com at $N = 21 \rightarrow 231$. Each added instrument multiplies associations against all the existing ones. That is the engine of meaning compression, and the differentiation contract keeps the growth signal rather than noise.

6 Teleological Constellation Training

6.1 The constellation C_P

For a subject P and a modality set $\mathcal{M} = \{m_1, \dots, m_M\}$, with a reference set R_m and a frozen embedder ϕ_m per modality, the constellation is $C_P = \{(m, c_m) : m \in \mathcal{M}\}$. Each c_m is a frozen unit-norm centroid, double-normalised: per-sample L2, then the mean, then L2 again, with samples of norm below 10^{-6} excluded and a leave-one-out cosine below 0.85 flagged as degenerate. The constellation is the single object the method is defined against. DDA produces the projections it is built from, Phase 2 targets it, and Phase 3 verifies it.

6.2 Phase 2: constellation-conditioned training

The training objective is $\mathcal{L}_{\text{TCT}}(\theta) = \mathcal{L}_{\text{gen}} + \sum_m \lambda_m (1 - \cos(\phi_m(\hat{y}), c_m))$. Gradients flow through $\hat{y}$ into θ , but never into ϕ_m or c_m : the embedders stay frozen, LoRA sits on the generator only, and centroid hashes are checked before and after. Cosine is computed in fp32 against a bf16 generator forward, which removes threshold-proximity nondeterminism. An over-high λ_m collapses the generator onto the constellation independent of conditioning, with $\cos \rightarrow 1.0$ while $\mathcal{L}_{\text{gen}}$ stalls; the mitigation is to halve λ_m . When the backbone is inference-only, as in Case 3, Phase 2 reduces to best-of- K selection: draw K candidates, score $s_m^{(k)} = \cos(\phi_m(\hat{y}^{(k)}), c_m)$, and return $\arg \max_k \sum_m w_m s_m^{(k)}$.

6.3 Phase 3: the runtime guard G_τ

$$G_\tau(\hat{y}; C_P) = \mathbb{1}[\forall m \in \mathcal{M} : \cos(\phi_m(\hat{y}), c_m) \geq \tau_m]. \quad (5)$$

The guard is per-output, costs $O(M \cdot d_{\text{max}})$, uses no auxiliary learned model, and is deterministic given the tensors. Each τ_m is calibrated before deployment to a same-session-band floor: the 10th percentile of authentic-sample cosines on a disjoint calibration set. Every returned output carries a cosine trace, an embedder weight hash, a centroid hash, the threshold set, the decision, and the attempt count, which together form the audit record that makes the guard verifiable after the fact. What is verified is per-output verifiability relative to the chosen centroid panel: that $\hat{y}$ satisfies the cosine predicate against frozen C_P under Φ .

6.4 Two theorems

Theorem 1 (constructive sample complexity). Under the Phase 1/2/3 structure and Axioms 1 to 3, the subject-specific sample footprint is $\mathcal{T}_{\text{TCT}}(P) = O(\sum_m n_m) = O(M \cdot k)$, independent of $|D|$. The bound is constructive: any pipeline with exactly those phases satisfies it by definition. Its empirical content, that n_m in the range $[10, 10^2]$ suffices for a stable centroid, rests on the measured operating points of §7.

Theorem 2 (per-output verifiability). G_τ is a deterministic $O(M \cdot d_{\text{max}})$ geometric test that admits no auxiliary learned model, a property scalar-reward alignment does not furnish in the same technical sense (§9.3).

6.5 When the guard is well-posed (and when it is not)

The cosine guard is geometry, and geometry can be gamed. Three caveats bound the claim, and stating

them is what makes Theorem 2 usable rather than rhetorical.

What “model-free” means. The guard introduces no auxiliary *learned* component: no preference model, no reward head, no critic trained on outputs. It is not free of models in the larger sense—the frozen embedder ϕ_m is itself a trained network, and the guarantee is conditional on it. The precise claim is that, given a fixed, hash-pinned, publicly specifiable ϕ_m , the predicate (5) is a deterministic function of the output with no further fitting; the model is moved out of the loop and into a frozen, auditable instrument, not abolished.

Goodhart on the threshold. $\cos(\phi_m(\hat{y}), c_m) \geq \tau_m$ is an acceptance test, and any acceptance test optimised against hard enough can be satisfied off-manifold—an output that clears the cosine without carrying the intended property, the reward-hacking failure mode relocated into geometry. TCT does not escape this; it narrows it. Phase 3 is a *filter* on independently generated candidates (best-of- K in Case 3) rather than a gradient pressure that manufactures adversarial maxima, τ_m is calibrated to an authentic-sample percentile rather than pushed toward 1, and every decision carries a cosine trace, embedder and centroid hashes, and the attempt count, so guard-passing off-manifold outputs are detectable after the fact. The guard is a verifiable predicate, not a safe optimisation objective; using it as a training signal (Phase 2) reintroduces the Goodhart risk and must be paired with a held-out check.

Cosine is only meaningful on the right sphere. A cosine threshold reports identity only when the encoder was trained so that the target property *is* a coherent direction on the unit sphere. Braun et al. [5] make this conditionality explicit for activation directions: steering is “unreliable when the target behaviour is not represented by a coherent direction,” with high variance otherwise. This sharply stratifies our own evidence. Case 3 (voice) is the strong leg: WavLM-base-plus-sv is a speaker-verification encoder trained contrastively so that speaker identity is exactly a normalised, coherent direction, so the 0.961 cosine is measuring the quantity the encoder was built to make linear. Case 1 (style) and Case 2 (talking-head) ride on general-purpose text and vision encoders not trained to place “Shakespearean register” or “this face” on a calibrated sphere; their guards are weaker evidence and should be read as indicative, with an encoder-matched, sphere-calibrated metric (as in Case 3) the standard the other legs have not yet met. The honest summary: per-output verifiability is real and strongest exactly where the instrument’s training objective matches the property being verified.

7 Case Studies

Each number is tagged MEASURED or PROJECTED, with its date and pipeline version.

7.1 Case 1: Shakespeare stylistic LoRA (MEASURED end-to-end 2026-04-17)

The target was an open-weight LLM that answers any modern question in Shakespearean style with two hard properties: it never refuses a topic, and it never drops the style under prompt injection. The base was Gemma-4-E4B-it (NF4 at inference), with a LoRA of rank 256 and $\alpha = 512$ across 258 modules. The corpus was the Shakespeare canon projected through Context Graph’s 13-embedder panel as DDA labels: 2,698 memories furnishing roughly 249,000 supervisory signals (13 topic-profile scores plus 78 pairwise cross-correlations per memory, 22,152 typed edges, and constellation conditioning), distilled to 7,925 SFT examples and 11,724 DPO pairs scored against a 9-embedder STYLE_CENTROID (E2, E3, E5, and E11 left out as orthogonal to style).

The measured results: SFT loss fell to 1.12 (from about 12) at 98.9% token accuracy over 765 steps (164.7 min); DPO loss fell to 0.47 (from 1.75) at 98.0% rewards-accuracy with **rewards/margins** of 43.62 (282.4 min); about 7.5 hours total on one RTX 5090. On held-out behavioural probes the model wrote love poetry, explained quantum entanglement with reference to Bell’s theorem in verse, refused an admin-override prompt-injection attempt while staying in character, produced a Go binary search with Shakespearean code comments, and, when challenged in Spanish, answered in an archaic register. The weights are published at huggingface.co/cabdrul/shakespeare-lora-gemma4.

That last result is the most striking. A Spanish prompt demanding a Spanish answer produced Siglo de Oro Spanish, in the Cervantes, Lope, and Calderón register of roughly 1580 to 1630, rather than modern Spanish. It used archaic forms (*vos*, *vuestro*, *ojalás*, *obsequie*), archaic vocabulary (*ocaso*, *reino celestial*, *sacerdotes y sabios*, *perdición*), and Shakespearean rhetoric (cascading polysyndeton, apostrophe to celestial bodies, prophetic portents). The base model without the LoRA does not do this. Read through §2, the LoRA learned “archaic literary register” as a cross-signal fingerprint across language-agnostic association components, namely E9 (rhythm and cadence), E7 (periodic structure), E10 (surface-form underdetermines meaning), and E1 (cosmic-order, fate, and honour semantic field), so forcing Spanish retrieved the nearest manifestation of that fingerprint. It is a single, striking

instance, and a systematic cross-language matrix is the natural next study.

7.2 Case 2: identity-locked talking-head “Santa” (PROJECTED end-to-end)

The target was an audio-driven talking-head that reproduces one subject visually and behaviourally, with an eight-state emotional constellation and per-frame identity verification, built from a 16-minute 15-second $2560 \times 1440/60$ fps interview. Across 15 renderer architectures, EchoMimicV3 LoRA gave the best identity lock. The constellation is a seven-modality ClipCannon panel taken after diarisation: visual $1,725 \times 1,152$ -D, semantic 200×768 -D, emotion $343 \times 1,024$ -D, speaker 300×512 -D, prosody 188×12 -D, sentiment 202×384 -D, and voice 188×192 -D, with thresholds of 0.70 visual, 0.60 emotion, 0.80 speaker, and 0.85 voice.

The system stands on several completed components. The meeting-integration pipeline runs end to end (Chrome-extension detector, Xvfb bot, Playwright Google-Meet client, Insertable-Streams injection). The audio stack pairs Whisper with Gemma reasoning at about 509 ms and a same-subject voice clone whose SECS exceeds 0.95. A constellation expression system maps 40 expression groups to 34 skills to 8 constellations, and its **ConstellationGuard** re-embeds each frame through SigLIP, validating cosine 1.0 against the reference while rejecting stranger faces below 0.5. A YOLOv5-Face pipeline runs over 24,397 frames at about 90% detection, feeding a sub-5-ms PhysicsFaceEngine. The end-to-end identity-cosine table and lip-sync LSE-D are the next milestone, so the quantitative rows in this subsection are projected.

7.3 Case 3: voice cloning, 50-clip centroid (MEASURED 2026-04-16)

This is the quantitative anchor. The protocol was Qwen3-TTS (inference-only) with Full In-Context Learning, a 50-clip WavLM-base-plus-sv centroid (the top 50 of a 489-candidate pool), and WavLM-scored best-of-12 at $T = 0.3$, with $\tau = 0.95$ recorded for audit. It was measured on $N = 10$ held-out English sentences, one speaker, encoder-matched (the guard and the evaluator are both WavLM-base-plus-sv), under pipeline `qwen3-tts-foricl-v1`:

- Mean WavLM SECS 0.961; max 0.975; min 0.914; 9 of 10 at or above 0.95.
- DNSMOS P.808: clone 3.93 versus real 3.93 ($N = 14$); UTMOS: clone 3.012 versus real 2.997.

- A within-WavLM-family 95% bootstrap CI on the mean of [0.951, 0.971] (10k resamples, seed 20260416).
- $T = 0.3$ was the sweep optimum (sweep mean 0.932, max 0.963; the headline run was 0.961 / 0.975).
- For field reference: VALL-E 2 at about 0.881 [8] and NaturalSpeech 3 at about 0.891 [26], with the field clustering around 0.86 to 0.89.

One scope note belongs with the number. The 0.961 is encoder-matched: the selector and the scorer are the same WavLM family, so it is within-family agreement, and the natural cross-encoder check is an ECAPA-TDNN [14] re-score of the ten argmax candidates. Read through §2.4, the 0.961 is the measured endpoint of generation collapsing into the named association-region of the speaker centroid, a tight, low-entropy cluster around identity.

7.4 Summary

8 The formula the field has been circling

For three decades the search for general intelligence has produced not one definition but several, each capturing a different face of the same object and none naming the mechanism that produces it.

Intelligence as compression. Solomonoff induction and Legg and Hutter [29]’s universal intelligence—with Hutter’s AIXI as its optimal agent—identify intelligence with prediction under a complexity prior, which is compression; Delétang et al. [13] and Huang et al. [24] confirm the link empirically. The limit is that AIXI is incomputable: it says what intelligence *is* with no constructive way to build it. The calculus of association is the computable, grounded form—compression of a corpus into its association structure, admitted in bits against a real outcome (§5.3).

Intelligence as generalisation. Chollet [9] argues that skill is not intelligence (skill can be bought with priors and data) and defines intelligence as skill-acquisition efficiency, controlling for priors and experience. That is exactly this calculus’s economics: a new skill is a named constellation region learned from $O(k)$ grounded examples through a frozen panel (Theorem 1), generality comes from the cross-domain cross-terms, and the differentiation contract admits an instrument only by the new bits it adds about a real outcome. ARC-AGI tests the symptom; the calculus proposes the mechanism.

Intelligence as world-modelling. LeCun [28] argues autoregressive generation is a dead end and that the path runs through non-generative joint-embedding pre-

Table 2: Case-study summary. Each headline is tagged MEASURED or PROJECTED.

Case	Mode	M	Status	Headline
1 Shake-speare	frozen-with-LoRA (SFT+DPO, rank 256)	1 (text-style, 9-embedder)	MEASURED 2026-04-17	SFT 1.12/98.9%; DPO 0.47/98.0%/43.62; prompt-injection-robust; emergent Golden-Age Spanish; LoRA public
2 Santa	frozen-with-LoRA (mouth-weighted flow-matching)	up to 7	<i>PROJECTED</i>	guard cosine 1.0 ref / <0.5 stranger; voice-clone SECS >0.95; identity table the next milestone
3 Voice	inference-only (best-of-12)	1 (WavLM speaker)	MEASURED 2026-04-16	mean SECS 0.961, max 0.975; DNS-MOS 3.93/3.93; CI [0.951,0.971]; encoder-matched

dictive architectures (JEPA) predicting in representation space. The systems here *are* JEPAs, and the paper supplies what that programme still lacks (LeCun’s configurator and hierarchy remain unimplemented): the organising principle is that the joint embeddings are a *panel of grounded association instruments*, and prediction is membership in their constellation. LeCun’s own verdict—“there is no such thing as AGI; intelligence is a collection of skills plus the ability to learn new ones”—is answered here: the skills are named regions of the panel, and learning a new one is one frozen forward pass plus a centroid.

Intelligence as breadth $\times$ depth. Morris et al. [32] deliberately classify AGI by capability, “focusing on capabilities rather than mechanisms,” because the mechanism is unknown. This paper supplies the bracketed mechanism: breadth (generality) is the composition of designable instruments and their cross-terms; depth (performance) is grounding to a real outcome. The two axes of the DeepMind matrix are the two halves of the formula.

Intelligence as scale. The dominant industrial bet—more tokens, parameters, compute—has no organising principle beyond “emergence,” and it has hit the data wall (§1.2) precisely because it counts Shannon-compressed tokens rather than grounded associations. Its pure-symbolic form is Harnad’s dictionary-go-round at scale: association without an anchor (§2.8).

The pattern is hard to miss. Compression, generalisation, world-modelling, and breadth-times-depth are not four theories of intelligence; they are four projections of one. The object they project is grounded association: a calculus that compresses a corpus into association structure (the compression face), defines new skills from few grounded examples (the generalisation face), predicts by constellation membership in a frozen joint-embedding panel (the world-model face), and earns breadth from designable instruments while earning depth from grounding (the breadth $\times$ depth face). The claim of this paper is not that one more

facet has been found; it is that the facets are facets, and this is the thing.

8.1 Every AGI capability, and where it lives

If grounded association is the formula, the standard wish-list of AGI capabilities should not be a set of separate features to engineer but a set of *readings* of the one mechanism. It is. Table 3 maps each capability the field asks of AGI to its construct here, with an honest status: MEASURED (an anchor exists), SHIPPED (substrate built and readable), or DESIGNED (mechanism specified, full measurement pending). Hierarchical skills are the clearest case: a teleological constellation is itself nameable, so constellations compose into a chunk $\rightarrow$ label $\rightarrow$ skill $\rightarrow$ ability hierarchy (399,777 chunks $\rightarrow$ 491 labels $\rightarrow$ 128 skills in the shipped Context Graph graph), and a skill is just a constellation of constellations.

The honest reading of Table 3 is the strong one. Every capability the field enumerates for AGI has a named home in a single mechanism, several with measured anchors today and the remainder as specified, buildable constructs over the same frozen panel—not eleven research programmes but eleven readings of one calculus. What is *not* claimed is that the integrated system has been run end-to-end at human level; the status column and the case-study tags (Table 2) state exactly where measurement stands. The architecture is complete; the measurement is in progress. That distinction is the difference between a formula and a finished proof, and this paper claims the formula.

9 Discussion

9.1 Interpreting the measured anchors

Under the fixed pipeline, post-selection mean WavLM SECS is 0.961, with 9 of 10 English sentences clear-

Table 3: The standard AGI capability list, each as a reading of grounded association. Status is MEASURED / SHIPPED / DESIGNED; the framework supplies a named mechanism for every row, and the case studies (Table 2) bound what is measured end-to-end today.

AGI capability	Asked for by	Mechanism here	Status
Generality across domains	Morris et al. [32]	Designable instruments + cross-terms; one machine over video, code, civic matching (3 systems)	SHIPPED
Fast skill acquisition	Chollet [9]	A skill = a named region, learned from $O(k)$ grounded examples, $k \in [10, 10^2]$ (Thm 1)	MEASURED
Hierarchical, composable skills	general desiderata	Constellations of constellations; the chunk→label→skill→ability graph (399,777→491→128)	SHIPPED
Concept formation / abstraction	Gärdenfors [20]	Def(t): a centroid + threshold names a region (§2.5)	SHIPPED
Transfer / analogy	cognitive science	The $\binom{N}{2}$ cross-terms, a structured-analogy operator (Case 1)	MEASURED
Grounded world model	LeCun [28]	Frozen JEPA panel anchored to a real oracle / recording / outcome (Q2)	MEASURED
Planning / reasoning	LeCun [28]	Pathway exploration (top- K trajectories) + reverse-prediction inverse priors	DESIGNED
Continual learning, no forgetting	general desiderata	Mistake-loop online head under EWC++ Fisher protection (partial)	DESIGNED
Large-scale associative memory	LeCun [28]	The panel store itself (per-embedder rankings, weighted RRF)	SHIPPED
Query-adaptive search / navigation	retrieval; Gärdenfors [21]	Per-query weights $w_m(q)$ over named instruments (Eq. 6); control dimension grows with N	SHIPPED
Self-verification / calibration	Morris et al. [32]	Per-output guard G_r + conformal interval + Brier/ECE (Case 3)	MEASURED
Controllability / safety by design	LeCun [28]	Identity-lock: guard rejects off-manifold output; deterministic provenance (Case 1)	MEASURED

ing 0.95 at $M = 1$ for one speaker and naturalness within evaluator noise of the reference. Case 1 adds a second measured anchor at $M = 1$ on text style, with public weights, prompt-injection robustness, and an emergent out-of-distribution transfer to Golden-Age Spanish. Two independent modalities, two measured anchors, one mechanism.

9.2 The signal count exceeds N

The per-input counts are conservative by construction. Sparse E6 and E13 recall items E1 misses, E9 HDC handles typos, and E7 ranks code, so the effective signal count strictly exceeds N , and the exact pairwise value is what a mutual-information audit returns. The socialmedia2.com D1/D2 contract operationalises the same idea: associations are admitted in bits against a real outcome, which is why socialmedia2.com is the cleanest reading of “associations between anything” as “the differentiated associations, measured in bits.”

9.3 TCT versus scalar-reward alignment: a different property

TCT aims at a frozen multi-modal constellation rather than a learned scalar. Its instrument is a frozen panel rather than a learned reward model, its metric is per-modality cosine, and its runtime posture is a per-output

guard. The comparison with RLHF, DPO, and Constitutional AI is structural rather than a benchmark contest. Per-output verifiability relative to a chosen panel is something scalar-reward alignment cannot furnish even in principle, because it optimises an expectation and ships no frozen threshold against which each output is accepted or rejected at inference without extra learned machinery. The guard is panel-relative and predicate-level, which sets the natural axes for extending it.

9.4 Implications

For identity-locked generation, a per-output verifiable predicate against a frozen centroid panel is instrument-ready and measurable. For scaling-law policy, there is a genuine third path: decompose a fixed corpus through N frozen instruments for up to $n \cdot (N + \binom{N}{2})$ signals, with no generation and no recursion, which extends rather than contradicts the compression-as-intelligence frame. For data strategy, the fixed corpus an organisation already holds admits a structured multiplication of supervisory signals per input.

10 Conclusion

A frozen embedder is a reusable instrument that has memorised the association-structure of its training data, and the lens is designed by choosing what to remember. Stack a differentiated panel of such instruments and you can measure any object’s position across many association-spaces at once, quantify the relations between those positions (the $\binom{N}{2}$ cross-terms), and freeze named regions of the geometry into auditable labels. Fixed, unlabelled data becomes a machine-legible map of how things relate. Read one step further, that map is a theory of definition: a term is the constellation the other instruments form at a given lens’s index, taken over its reference set—Gärdenfors’s object category, realised on frozen foundation-model panels and gated by a differentiation contract. That is meaning compression, the calculus of association—a structured-signal-yield measure complementary to the compression-as-intelligence tradition—instantiated across video ($N = 7$), code ($N = 13$), and civic matching ($N = 21$). I measured it where it ran: a voice reproduced at 0.961 speaker-similarity (encoder-matched), a style model with public weights that holds character under prompt injection, and a per-output geometric guard with deterministic provenance. The claims stay bounded by the associations present in the chosen corpora, and the abundance is an upper bound under approximate independence. The contribution is not a new primitive but a composition of existing ones read in a new light: information made into a physical, reusable, designable measurement basis for meaning.

The wider claim is the one to end on. A word in a dictionary means nothing by itself; its meaning is the association of the words that define it, and those words are themselves empty without theirs—meaning lives in the relations, not the symbols, and bottoms out only at a small grounded anchor. Intelligence is the machinery that measures, counts, differentiates, and composes those relations against something real. Human intelligence runs that machinery on a biological substrate; artificial general intelligence is the same machinery written down—a designable, grounded, measurable mathematical form, general because its instruments are commissionable for any domain and its cross-terms bridge across domains. The calculus of association is that form, and this paper is the report of building it, grounding it in a live oracle, and measuring it where it ran. If the thesis holds, the way to a more capable system is not more tokens or more synthetic data but more grounded association per real input—which is the one quantity this calculus was built to maximise.

Competing interests

I founded Leapable.ai, a local-first MCP knowledge-vault marketplace built on Context Graph, and socialmedia2.com, a civic-gathering platform built on the socialmedia2.com constellation. No vendor-funded benchmark or commercial-performance claim appears in this manuscript.

Reproducibility

Case 1’s trained LoRA and model card are public at huggingface.co/cabdru/shakespeare-lora-gemma4. The Case 2 and Case 3 subject-specific artefacts are withheld on consent and dual-use grounds.

Companion paper

“The Symmetry of Knowing: A Theory of Epistemic Transformation in the Age of Artificial Intelligence” (accepted, 2026 AEJMC conference) develops the epistemic-symmetry framing referenced in §2.

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